MAT2002 **Name (Print):** _____
Spring 2026
HW4
Deadline: 12th April

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1. Verify that the given functions are solutions of the differential equation, and determine their Wronskian.
$$xy''' - y'' = 0; \quad 1, \quad x, \quad x^3$$

2. In each of following, find the general solution of the given differential equation.
   (i) $y''' - 5y'' + 3y' + y = 0$
   (ii) $y''' + 5y'' + 6y' + 2y = 0$

3. In each of following, determine the general solution of the given differential equation.
   (i) $y^{(4)} - 4y'' = t^2 + e^t$
   (ii) $y^{(4)} + y''' = \sin 2t$

4. In each of following, find the general solution of the given differential equation. Leave your answer in terms of one or more integrals.
   (i) $y''' - y'' + y' - y = \sec t, \quad -\pi/2 < t < \pi/2$
   (ii) $y''' - y' = \csc t, \quad 0 < t < \pi$

5. Given that x, x^2, and $1/x$ are solutions of the homogeneous equation corresponding to
$$x^3y''' + x^2y'' - 2xy' + 2y = 2x^4, \quad x > 0,$$
   determine a particular solution.

6. Transform the given initial value problem into an initial value problem for two first order equations.
$$u'' + p(t)u' + q(t)u = g(t), \quad u(0) = u_0, \quad u'(0) = u_0'$$

7. Solve the following systems or the initial value problem:
   (i) $\boldsymbol{x}' = \begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} \boldsymbol{x}$,
   (ii) $\boldsymbol{x}' = \begin{bmatrix} 2 & -1 \\ 1 & 4 \end{bmatrix} \boldsymbol{x}$;
   (iii) $\boldsymbol{x}' = \begin{bmatrix} 1 & -5 \\ 1 & -3 \end{bmatrix} \boldsymbol{x}, \boldsymbol{x}(0) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$.